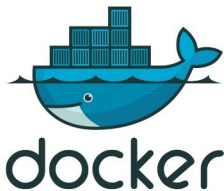


Run Docker on the Google Cloud Platform

Google Cloud Platform is one of the leading public cloud infrastructures that allows you to run applications, perform data analysis and various tools. Docker, on the other hand, is an engine that runs container workloads. This article shows the reader how to run Docker workloads on the Google Cloud Platform.



Organisations are moving their workloads to the cloud at a fast pace. At the same time, applications have been shifting towards modularity through microservices and the more efficient running of workloads, whether they are on premise or in the cloud. It is not just virtual machines (VMs) that organisations are provisioning in the cloud, but container runtimes within these VMs for higher efficiency. Docker has been at the forefront of all this, as the engine that runs these container workloads.

The Google Cloud Platform (GCP) provides multiple ways to run Docker workloads in the cloud. The options range from spinning up one's own VMs to fully managed container orchestration platforms (based on Kubernetes) that are offered as options to the customer. This is coupled with powerful APIs and command-line interfaces to provision and manage the infrastructure and applications in the Google Cloud.

Google Cloud Platform (GCP) provides multiple options, depending on how much infrastructure management is desired (for simple deployment of applications, Google can take care of managing the infrastructure for you). Let us go through the options, first in brief and then in detail.

Google Compute Engine: This is the Infrastructure as a Service (IaaS) offering from the Google Cloud Platform. Google provides multiple types of VMs with pricing options,

as well as the storage and networking infrastructure that might be needed for setting them up.

Google Container Engine: This is a fully managed offering that provides container orchestration at scale, based on Kubernetes. If you are dealing with Docker images, you can also choose to use the Google Container Registry option to host your images. For those fully committed to Google Cloud, using the Google Container Registry is a good option for faster availability of images.

Google App Engine Flexible: Building on the success of Google's Platform-as-a-Service (PaaS) offering, i.e., App Engine Standard, the Flexible Environment supports multiple other languages including bringing your own stack. The development workflow is based on Docker and eventual deployment is done in the Container Engine offering, behind the scenes, by the provisioning infrastructure.

All of the above services are available via the Google Cloud SDK, which is a set of command line utilities that can be used on a laptop to work with GCP resources. Those who have worked with Amazon Web Services (AWS) and Azure will find it familiar, as Google uses similar client tools to manage its cloud resources.

Google Cloud SDK

Google Cloud SDK is a suite of command-line tools for the platform. These include gcloud, bqutil, gsutil and more